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Production Analysis of Linear Flow Into Fractured Tight Gas Wells

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Abstract

Sometimes decline curves for tight gas wells indicate that linear flow may last for over 10 or 20 years. These decline curves may show outer boundary effects but *no pseudo-radial flow*. This paper presents decline curve analysis methods for such wells. Values for $\sqrt{k}\,x_f$ and drainage area can be calculated. Stabilized (bounded) flow equations are also presented for forecasting. The solutions and type curves developed are for both infinite-acting and stabilized flow with either constant p_{wf} or constant rate. Several field cases are shown with example calculations.

Introduction

Many gas wells have been observed which stay in the linear flow regime for several years. These are usually wells in very tight gas reservoirs which may have hydraulic fractures designed to extend to or nearly to the drainage boundary of the well. For these wells, no pseudo-radial flow is expected nor observed during the production period. Linear flow seems to be the dominant flow regime throughout the well's production life.

Fig. 1 is a top view of a hydraulically fractured well whose fracture extends all the way to the lateral boundaries. The well is in the center of a rectangular drainage area. The distance to the outer boundary in the direction perpendicular to the fracture is y_e. The fracture is assumed to have infinite conductivity. This is a good assumption for large dimensionless fracture conductivity, $F_{CD} > 50$.

Dimensionless fracture conductivity is defined by:

$$F_{CD} = \frac{k_f w}{k x_f} \qquad (1)$$

With these conditions, the flow is linear and is perpendicular to the fracture.

The petroleum literature has many complicated cases for hydraulically fractured wells[1-9]. Solutions for these cases are presented as type curves, tables, or sometimes as equations.

Usually the cases involve fractures in infinite reservoirs, which means that flow eventually leaves the linear regime and becomes "pseudo-radial". Almost all of the type curves are for constant rate flow. There are very few solutions for constant (wellbore) pressure, which is of interest for long-term production analysis.

Apparently, the only solution published for the problem of Fig. 1 is for a square with constant flow rate[9]. This is shown as a special case of a more general set of solutions.

Miller[10] provided solutions for linear flow in aquifers. Miller's solutions were for the infinite acting and bounded aquifers for both constant rate and constant pressure cases. Nabor and Barham[11] generalized these solutions in dimensionless variables. They also added the solutions for constant pressure outer boundary case. The mathematics for these solutions is also available from Carslaw and Jaeger[12] for linear heat conduction.

Wattenbarger, El-Banbi, and Maggard[13] have adapted these linear reservoirs solutions to fractured wells in the geometry of Fig. 1. They included the constant rate and constant pressure cases for linear flow in a rectangle. The equations and type curves in the following sections are new, for the most part, and provide tools for analyzing long-term production performance of tight gas wells.

Solutions of Linear Flow Into Fractured Wells

The following equations apply for linear flow into a fracture. These equations strictly apply to the liquid case, but they can be used for gas wells with good accuracy if dimensionless real gas pseudo-pressure[14], m_D, (for gas) is substituted for dimensionless pressure, p_D, (for liquid).

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We show the solutions for two inner boundary conditions (constant rate and constant p_{wf}) and closed reservoir outer boundary condition. Eq. 2 is the solution for constant rate production from a closed linear reservoirs.

$$p_{wD} = \frac{\pi}{2}\left(\frac{y_e}{x_f}\right)\left[\frac{1}{3}+\left(\frac{x_f}{y_e}\right)^2 t_{Dx_f}\right]$$

$$-\frac{2}{\pi^2}\left(\frac{y_e}{x_f}\right)\sum_{n=1}^{\infty}\left(\frac{1}{n^2}\right)\exp\left[-n^2\pi^2\left(\frac{x_f}{y_e}\right)^2 t_{Dx_f}\right] \quad \text{...(2)}$$

Eq. 3 is the solution for constant p_{wf} production from a closed linear reservoir.

$$\frac{1}{q_D} = \frac{\frac{\pi}{4}\left(\frac{y_e}{x_f}\right)}{\sum_{n_{odd}}^{\infty}\exp\left[-\frac{n^2\pi^2}{4}\left(\frac{x_f}{y_e}\right)^2 t_{Dx_f}\right]} \quad \text{.....(3)}$$

where dimensionless variables are defined by:

$$p_{wD} = \frac{kh(p_i - p_{wf})}{141.2qB\mu} \quad \text{.....(4)}$$

$$\frac{1}{q_D} = \frac{kh(p_i - p_{wf})}{141.2qB\mu} \quad \text{.....(5)}$$

$$t_{Dx_f} = \frac{0.00633kt}{\phi\mu c_t x_f^2} \quad \text{.....(6)}$$

Note that the definitions of p_{wD} and $1/q_D$ appear to be the same except that p_{wf} varies in the p_{wD} case while q varies in the $1/q_D$ case.

"Short-term" Approximations. The solutions given by Eqs. 2 and 3 have short-term approximations. These approximations are also the solutions for infinite reservoir outer boundary. These approximations are given for constant rate and constant p_{wf} by the following equations, respectively:

$$p_{wD} = \sqrt{\pi t_{Dx_f}} \quad \text{.....(7)}$$

$$\frac{1}{q_D} = \frac{\pi}{2}\sqrt{\pi t_{Dx_f}} \quad \text{.....(8)}$$

Unlike infinite radial reservoir solutions[15], note that constant rate and constant pressure solutions for infinite linear reservoirs differ by a factor of $\pi/2$.

Long-term Approximations. Closed reservoir solutions can be approximated by the following equations for long times.

$$p_{wD} = \frac{\pi}{2}\left(\frac{x_f}{y_e}\right)t_{Dx_f} + \frac{\pi}{6}\left(\frac{y_e}{x_f}\right) \quad \text{.....(9)}$$

$$\frac{1}{q_D} = \frac{\pi}{4}\left(\frac{y_e}{x_f}\right)\exp\left[\frac{\pi^2}{4}\left(\frac{x_f}{y_e}\right)^2 t_{Dx_f}\right] \quad \text{.....(10)}$$

These equations would result in different expressions for productivity indexes. The following equations are the productivity index expressions for constant rate and constant p_{wf} production respectively:

$$J_{CR} = \frac{kh}{141.2B\mu\left[\frac{\pi}{6}\left(\frac{y_e}{x_f}\right)\right]} \quad \text{.....(11)}$$

$$J_{CP} = \frac{kh}{141.2B\mu\left[\frac{2}{\pi}\left(\frac{y_e}{x_f}\right)\right]} \quad \text{.....(12)}$$

We see that the ratio of productivity index for constant p_{wf} production to the productivity index for constant rate production is $\pi^2/12$. This difference may be important and the right equation should be used for long-term production forecast.

Type Curves. The most practical way to illustrate our solutions is through log-log type curves. **Fig. 2** is a type curve for each of the two cases for all times of interest. These curves

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have been plotted against t_{Dye} rather than t_{Dxf}. The "y axis" has been modified to (x_f/y_e) p_{wD} and (x_f/y_e) ($1/q_D$) rather than p_{wD} and $1/q_D$. This gives only one curve for each case, for any rectangular geometry, rather than families of curves. The following equations become the solutions for closed reservoirs for constant rate and constant p_{wf} respectively:

$$\left(\frac{x_f}{y_e}\right)p_{wD} = \frac{\pi}{2}\left[\frac{1}{3}+t_{Dy_e}\right]$$

$$-\frac{2}{\pi^2}\sum_{n=1}^{\infty}\left(\frac{1}{n^2}\right)\exp\left(-n^2\pi^2 t_{Dy_e}\right) \quad \text{(13)}$$

$$\left(\frac{x_f}{y_e}\right)\frac{1}{q_D} = \frac{\frac{\pi}{4}}{\sum_{n_{odd}}^{\infty}\exp\left[-\frac{n^2\pi^2}{4}t_{Dy_e}\right]} \quad \text{(14)}$$

The dimensionless time, t_{Dye}, is defined by:

$$t_{Dy_e} = \frac{0.00633kt}{\phi\mu c_t y_e^2} = \left(\frac{x_f}{y_e}\right)^2 t_{Dx_f} \quad \text{(15)}$$

It is significant that these type curves (log-log plot of difference in pressure or reciprocal of production rate versus time) bend upward when the outer boundary is "reached". This is the opposite of the downward bend which is seen when fractured wells tend toward pseudo-radial flow[5,15]. This shape difference may be important in analyzing field data. (It should be understood that this rule is reversed when we plot production rate, rather than reciprocal of rate, versus time).

Figs. 3 and **4** are type curves drawn using Eqs. 2 and 3 for different aspect ratios, (x_f / y_e). **Fig. 5** is a decline curve drawn for the constant p_{wf} solution given by Eq. 14.

End of Linear Flow (1/2 slope line). An important feature of the type curves is the end of the infinite acting linear flow (seen as the "half-slope"). This can be used to estimate reservoir size from field data. Once again, the two cases are different. The end of the half slope is t_{Dye} = 0.5 for the constant rate case but is t_{Dye} = 0.25 for the constant p_{wf} case (see Fig. 2). These values, 0.5 and 0.25, are taken to be values where the curves *visually* depart from straight lines, making them useful for the purpose of establishing minimum drainage area from field data (discussed later).

Application to Gas Wells

The solutions for liquid flow can be adapted to gas flow by using the real gas pseudo-pressure[14]. In the liquid solutions, p_{wD} is replaced by m_{wD} which is defined as follows:

$$m_{wD} = \frac{kh[m(p_i)-m(p_{wf})]}{1424q_gT} \quad \text{(16)}$$

$1/q_D$ has the following definition:

$$\frac{1}{q_D} = \frac{kh[m(p_i)-m(p_{wf})]}{1424q_gT} \quad \text{(17)}$$

where $m(p)$ is the real gas pseudo-pressure[14] defined by:

$$m(p) = 2\int_{p_o}^{p}\frac{p}{z\mu}dp \quad \text{(18)}$$

and the dimensionless time referenced to the distance to the reservoir boundary, y_e, is defined in terms of initial fluid properties as follows:

$$t_{Dye} = \frac{0.00633kt}{(\phi\mu c_t)_i y_e^2} \quad \text{(19)}$$

The liquid solutions, which have been previously shown and are plotted in the type curves, apply very accurately for gas in transient flow. This has been determined for a number of well flow problems over the years. Thus, the use of Eqs. 16-19 works very well for the infinite-acting $\sqrt{t}$ behavior. Fraim and Wattenbarger normalized time[16] should be used to correct for changes in gas properties for boundary dominated flow data, if these solutions are used in forecasting. An alternative procedure for forecasting boundary dominated flow production will be discussed later.

Analysis of Field Data

With the above solutions, there are several ways to analyze gas well field data. The following is a description of analysis methods. An example well will be used later for illustrative purposes.

Log-Log Plot. The first step is to plot either pseudo-pressure difference versus time ($m(p_i)$- $m(p_{wf})$ vs. t), for constant rate

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production, or reciprocal of production rate versus time ($1/q_g$ vs. t), for constant p_{wf} production, on a log-log plot. This is a simple plot and will show indications of linear flow if the plot has a "half-slope".

Many wells may have periods where the well pressure varies or the well is shut in because of market curtailment, etc. For these reasons, the half-slope may not always be apparent or steady for the life of the well, even if the behavior of the well is entirely linear flow. Therefore, the log-log plot should be considered to be a screening method rather than a quantitative analysis tool.

Square-Root of Time Plot. The next step is to make a more definitive plot for linear flow - the $\sqrt{t}$ plot. The half-slope part of a log-log plot is equivalent to a straight line on the pseudo-pressure difference vs. $\sqrt{t}$ plot, for the constant rate case, or $1/q_g$ vs. $\sqrt{t}$ plot, for the constant pressure case.

The slope of this line is given by the following equations for constant rate and constant p_{wf}, respectively:

$$m_{CR} = \frac{200.8T}{h\sqrt{(\phi\mu c_t)_i}} \frac{q_g}{\sqrt{k}x_f} \quad \text{.....(20)}$$

$$m_{CP} = \frac{315.4T}{h\sqrt{(\phi\mu c_t)_i}} \frac{1}{\Delta m(p)\sqrt{k}x_f} \quad \text{.....(21)}$$

where:

$$\Delta m(p) = m(p_i) - m(p_{wf}) \quad \text{.....(22)}$$

Calculation of $\sqrt{k}\,x_f$. We can calculate $\sqrt{k}\,x_f$ from the slope of the $\sqrt{t}$ plot, m_{CR} or m_{CP}. From this slope, and knowing values for other reservoir parameters, we can use the following equations for constant rate and constant p_{wf} production, respectively:

$$\sqrt{k}x_f = \frac{200.8T}{h\sqrt{(\phi\mu c_t)_i}} \frac{q_g}{m_{CR}} \quad \text{.....(23)}$$

$$\sqrt{k}x_f = \frac{315.4T}{h\sqrt{(\phi\mu c_t)_i}} \frac{1}{\Delta m(p)m_{CP}} \quad \text{.....(24)}$$

This equation requires that permeability be known to determine x_f. Unless k is known independently, it is very difficult to determine x_f.

It should be emphasized that these two equations differ by $\pi/2$.

Calculation of "Distance of Investigation". The distance, y_e, can be calculated by identifying the end of the "half-slope" line, t_{ehs}, and comparing this time, in days, to the corresponding dimensionless time, t_{Dye}. This value of $(t_{Dye})_{ehs}$ is 0.5 for the constant rate case and 0.25 for the constant p_{wf} case. The corresponding distance to the outer boundary is then given by the following equations for both cases respectively:

$$y_e = 0.113\sqrt{\frac{kt_{ehs}}{(\phi\mu c_t)_i}} \quad \text{.....(25)}$$

$$y_e = 0.159\sqrt{\frac{kt_{ehs}}{(\phi\mu c_t)_i}} \quad \text{.....(26)}$$

This boundary distance can be considered to be *minimum* value if all the history data is still on the half-slope trend. In this case, the latest production time is used instead of $(t)_{ehs}$ in Eqs. 25 and 26. This would be the "distance of investigation" at the current time. Again, these equations require that permeability be known. Of course, this may be a weak point in determining y_e by this method since k may be uncertain.

Calculation of drainage area. The good news is that the drainage area can be calculated directly, without knowing either k or x_f. For the model of Fig. 1, the drainage area is given by:

$$A = 4\,x_f y_e \quad \text{.....(27)}$$

Eq. 27 can be used to calculate drainage area from the slope, m_{CR} or m_{CP}, and the end of half-slope time, t_{ehs}, for both cases:

$$A = 4x_f y_e = \frac{90.8T}{h(\phi\mu c_t)_i}\left\{\frac{q_g}{m_{CR}}\right\}\sqrt{t_{ehs}} \quad \text{.....(28)}$$

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$$A=4x_f y_e=\frac{200.6T}{h(\phi\mu c_t)_i}\left\{\frac{1}{m_{CP}\Delta m(p)}\right\}\sqrt{t_{ehs}} \quad \text{.....(29)}$$

where the drainage area, A, is in ft^2 but can be changed to acres by dividing by 43,560. Once again, if the well performance is still in the half-slope period, the latest time is used in Eqs. 28 and 29 and the drainage area is considered to be *minimum* (proved).

Calculation of Pore Volume. Now we can also calculate the pore volume, V_p. We get a direct determination of V_p if we multiply Eqs. 28 and 29 by ϕ and h. This may be a big help, since ϕ and h are often not known in tight gas formations. The equation for V_p for the constant rate is given by:

$$V_p=\frac{90.8T}{(\mu c_t)_i}\left\{\frac{q_g}{m_{CR}}\right\}\sqrt{t_{ehs}} \quad \text{.....(30)}$$

and for the constant p_{wf} case by:

$$V_p=\frac{200.6T}{(\mu c_t)_i}\left\{\frac{1}{m_{CP}\Delta m(p)}\right\}\sqrt{t_{ehs}} \quad \text{.....(31)}$$

We notice that the calculation of V_p does not depend on having good estimates of k, h, and ϕ.

Calculation of *OGIP*. Once the pore volume is calculated, then *OGIP* can be directly calculated by the following equation:

$$OGIP=\frac{V_p(1-S_w)}{B_{gi}} \quad \text{.....(32)}$$

If gas compressibility dominates c_t, the calculation of *OGIP* becomes insensitive to the value used for S_w. Thus, we may determine *OGIP* accurately without actually having good knowledge of k, h, ϕ, and S_w.

Example Calculation

We choose to apply our linear flow analysis techniques to production data of a hydraulically fractured tight gas well in a South Texas field. We used monthly rates for our plots. There were no recorded pressures in the early years, but it was assumed that pressure was constant throughout the producing time of the well.

Fig. 6 shows a log-log decline curve for the example well. The production rate follows a "half-slope" pattern for most of its life. This is an indication of linear flow with constant bottom-hole pressure. Actually, it is seen that the data starts bending downward after about fifteen years. Applying our linear flow analysis indicates that the well encountered the outer boundary at y_e at this time. Note that this well is not tending toward pseudo-radial flow since that would make the curve bend upward from the half-slope, not downward.

For this well, we need further analysis and finally confirmation by reservoir simulation since the variation of production rates and possible variation of pressures must have some effect on the decline curve. Nonetheless, the log-log decline curve seems to be a useful and easy analysis tool for our example well.

Fig. 7 shows the $1/q_g$ vs. $\sqrt{t}$ plot for the example well for a period of nearly 23 years. The cumulative production is also a straight line when plotted vs. $\sqrt{t}$ (not shown in the paper). The slope of the cumulative curve is $2/m_{CP}$. This tends to smooth the data, but subtleties in the boundary effect may not be seen as easily.

This figure was used to obtain both the slope, m_{CP}, and end of half-slope time, t_{ehs}. The values are 0.00009 and 5,625 respectively. Other well data are given in **Table 1**. The results of the calculations from Eqs. 24, 29, 31, and 32 are as follows:

$$\sqrt{k}x_f = 26.2 \text{ md}^{1/2}\text{ ft.}$$

A = 2,872,000 ft^2 = 65.9 acres

$V_p = 3.96 \times 10^7$ rcf

$OGIP = 6.67 \times 10^9$ scf = 6.67 Bcf

The above analysis should be accurate if all the assumed conditions are perfectly met. In the example well as well as many other wells in the same field, the reservoir/fracture system seems to cause linear flow; no pseudo-radial flow was detected. However, the conditions of constant rate or constant p_{wf} were never perfect. In some cases, the actual flowing conditions were not even close to either case. Therefore, it was necessary to confirm or modify the analysis by history matching with GASSIM[17], a single phase gas simulator. The simulation results were close to the results we obtained with the simple hand calculation technique.

Discussion

The analysis presented here may be useful for many fractured tight gas wells. This method based on linear flow analysis has not previously appeared in the literature although much of the mathematics was previously available. We expect that there will be increasing use of long hydraulic fractures in the future and closer well spacing as well. These two will tend to fulfill the conditions of this linear flow analysis method.

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The analysis here is simple but may have wide application. It is worthwhile to discuss several aspects of its application.

In a recent analysis of a field with about 60 wells, long-term linear flow was observed in about one-third of the wells. Many of the other wells could not be analyzed because of severe rate fluctuations caused by gas market problems. But none of the wells showed pseudo-radial flow theoretically expected in wells with hydraulic fractures.

Finite Conductivity Fractures. This analysis has been for infinite fracture conductivity. When formation permeability is extremely low, the conductivity of the fracture will behave as though it is relatively high in most cases. If the fracture conductivity is significantly low (F_{CD} less than 50) then this would appear as a "skin" in long-term production behavior. This type of behavior was analyzed by Agarwal, *et al*[7]. This would be recognizable by a non-zero intercept on the y axis of the $1/q_g$ vs. $\sqrt{t}$ plot.

This intercept was not noticed on any of the wells in the field analyzed. It should be recognized, however, that the early behavior was sometimes interrupted by the problems of putting a new well on production. It is interesting that pressure buildup tests on the example well have shown early flow to be dominated by bi-linear flow, indicating that the fracture conductivity was limited. Still, this was not noticeable on the long-term production behavior.

Possible Causes of Linear Flow Behavior. In Fig. 1 the fracture length extends all the way to the drainage boundary ($x_f = x_e$). This may not seem to be plausible in some cases. However, there may be other reasons for seeing this type of linear behavior.

Stright and Gordon[18] reported long-term linear behavior in tight gas wells which did not have particularly large fracture treatments. They believed that this might be caused by natural fracturing in the formation caused by normal tectonic processes in a relatively hard formation. These fractures would tend to be parallel to the fracture plane and would promote linear flow even if the fracture length were limited.

Fig. 8 shows how directional fractures in a reservoir would tend to change the reservoir into an equivalent "long, skinny" reservoir. The sketch on the left shows natural fractures oriented in the x direction. After the transformation which accounts for the anisotropic permeability, the dimensions in the x direction are changed to:

$$X = x\sqrt{k_y / k_x} \quad \text{(33)}$$

while the dimensions in the y direction are changed to:

$$Y = y\sqrt{k_x / k_y} \quad \text{(34)}$$

It is likely that anisotropy is sometimes much more severe than this example.

The calculated $\sqrt{k}\, x_f$ from either Eq. 23 or 24 would be the product of the square root of the equivalent permeability given by Eq. 35 and the transformed dimension X_f.

$$K = \sqrt{k_x k_y} \quad \text{(35)}$$

However, the calculated drainage area, A, and consequently the pore volume, V_p, and the *OGIP* are correctly determined even with the anisotropy.

Kohlhaas and Abbott[19] reported another possible reason for linear flow behavior other than the geometry of Fig. 1. They described massive tight formations which could be expected to have layers of higher permeability. The pressure drops in the higher permeability layers would then cause *vertical* linear flow into the higher permeability layers. This, too, would result in long-term linear behavior.

Layered Reservoirs at Linear Flow. The analysis given here is for a single layer reservoir. If the well produces from multiple layers, it would still appear as linear flow if flow is "infinite acting" in all layers. The calculated $\sqrt{k}\, x_f$ product would be a thickness weighted average summation of these values for the individual layers as shown by the following equation:

$$\sqrt{k}\, x_f = \frac{\sum_{j=1}^{n} \sqrt{k_j}\, x_{fj}}{\sum_{j=1}^{n} h_j} \quad \text{(36)}$$

Determination of Fracture Length. This is a continual puzzle for a fractured well. In our fractured well analysis, it is seen that the $\sqrt{k}\, x_f$ product is determined from the well behavior. The conventional thought is that permeability must be determined from a pre-fracture buildup test. This would work in a simple reservoir, but it is expected that many tight gas reservoirs will behave like multiple-layered reservoirs. In these cases, the permeability averaging that takes place before the fracture treatment and the $\sqrt{k}\, x_f$ averaging that takes place after the fracture treatment may not be related. Some type curves of complicated cases may help this problem but the results may not be unique.

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Forecasting. Once *OGIP* has been determined, forecasting is straightforward. Use the most recent data which has a reliable value of stabilized rate and bottom-hole pressure. From this, a productivity index can be determined directly, based on the following equation:

$$J_g = \frac{\left[m(\bar{p}) - m(p_{wf})\right]}{q_g} \quad \text{.....(37)}$$

If the boundary effects have not yet been observed, this will be a conservative forecast. The average pressure, $\bar{p}$, is determined from material balance, using actual cumulative production. Forecasting is then just a matter of selecting time intervals and applying material balance and productivity index equations at each future time interval.

Conclusions

1. Many wells in tight gas reservoirs have long-term production which exhibits only linear flow. These wells do not show pseudo-radial flow as sometimes expected with hydraulic fractures.
2. The equations for analyzing long-term (constant p_{wf}) linear flow production are different than for buildup analysis (constant rate). Both sets of equations are presented here.
3. Drainage area can be directly determined if the outer boundary effect has been observed. (If the reservoir is still infinite-acting, this would be a minimum value). Knowledge of permeability, k, is not required.
4. Pore volume and *OGIP* can be directly determined if the outer boundary effect has been observed. (If the reservoir is still infinite-acting, these would be minimum values). Knowledge of permeability, k, thickness, h, and porosity, ϕ, are not required.
5. If gas compressibility dominates c_t, the calculation of *OGIP* becomes insensitive to the value used for S_w. Thus, we may determine *OGIP* accurately without actually having good knowledge of k, h, ϕ, and S_w.

Nomenclature

A = well drainage area, L^2, ft^2.
B = oil FVF, dimensionless, RB/STB
B_g = gas FVF, dimensionless, rcf/scf
B_{gi} = gas FVF at initial pressure, dimensionless, rcf/scf
c_g = gas compressibility, Lt^2/m, psa^{-1}
c_t = total compressibility, Lt^2/m, psa^{-1}
F_{CD} = dimensionless fracture conductivity
FVF = formation volume factor, dimensionless
h = net formation thickness, L, ft.
J_{CP} = productivity index for constant p_{wf} production, L^2/mt^3, STB/D/psi
J_{CR} = productivity index for constant rate production, L^2/mt^3, STB/D/psi
J_g = gas productivity index, L^4t^2/m, Mscf.cp/D/psi^2.
k = permeability, L^2, md
k_f = fracture permeability, L^2, md
k_x = permeability in the X-direction, L^2, md
k_y = permeability in the Y-direction, L^2, md
K = equivalent permeability for anisotropic reservoir, L^2, md
m_{CP} = slope of $1/q_g$ vs. $\sqrt{t}$, $D^{1/2}$/Mscf
m_{CR} = slope of Δm vs. $\sqrt{t}$, psi^2/cp $D^{1/2}$
$m(p)$ = real gas pseudopressure, m/Lt^3, $psia^2$/cp
$m(\bar{p})$ = $m(p)$ at average reservoir pressure, m/Lt^3, $psia^2$/cp
$m(p_{wf})$ = $m(p)$ at flowing wellbore pressure, m/Lt^3, psi^2/cp
n = number of layers in a linear layered reservoir
$OGIP$ = Original Gas in Place, m^3, scf
p = absolute pressure, m/Lt^2, psia
$\bar{p}$ = average reservoir pressure, m/Lt^2, psia
p_D = dimensionless pressure
p_o = arbitrary lower limit of $m(p)$ integration, m/Lt^2, psia
p_{wf} = bottom-hole flowing pressure, m/Lt^2, psia
q_D = dimensionless flow rate
q_g = gas flow rate, L^3/t, Mscf/D
q = oil flow rate, L^3/t, STB/D
S_w = water saturation, fraction
t = producing time, days
t_{Dxf} = dimensionless time based on x_f
t_{Dye} = dimensionless time based on y_e
T = reservoir temperature, T, °R
V_p = pore volume, L^3, ft^3
w = fracture width, L, ft.
x_f = fracture half-length, L, ft.
x_e = reservoir half-width, L, ft.
X_e = transformed reservoir half-width because of anisotropy, L, ft.
y_e = distance from fracture to outer boundary, L, ft.
z = gas deviation factor, dimensionless
ϕ = porosity, fraction
μ = viscosity, m/Lt, cp
μ_g = gas viscosity, m/Lt, cp

Subscripts

ehs = end of "half-slope" period
i = initial conditions
j = layer index

Acknowledgments

We thank Tai Pham of Coastal Oil and Gas Corp. for sharing field data with us. We also thank the Reservoir Modeling Consortium for providing funding for this project.

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Table 1 - Data for example well.

Initial pressure, p_i	8800	psia
bottom-hole flowing pressure, p_{wf}	1600	psia
pseudo-pressure at p_i, $m(p_i)$	2.67×10^9	psi^2/cp
pseudo-pressure at p_{wf}, $m(p_{wf})$	1.69×10^8	psi^2/cp
gas specific gravity, γ_g	0.717	
reservoir temperature, T	290	°F
formation net pay thickness, h	92	ft.
formation porosity, ϕ	0.15	
average water saturation, S_w	0.47	
total compressibility at p_i, c_{ti}	3.53×10^{-5}	psi^{-1}

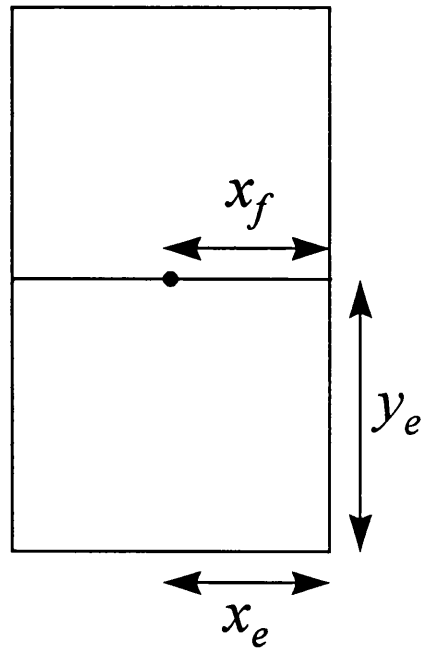

Fig. 1 - A hydraulically fractured well in a rectangular reservoir.

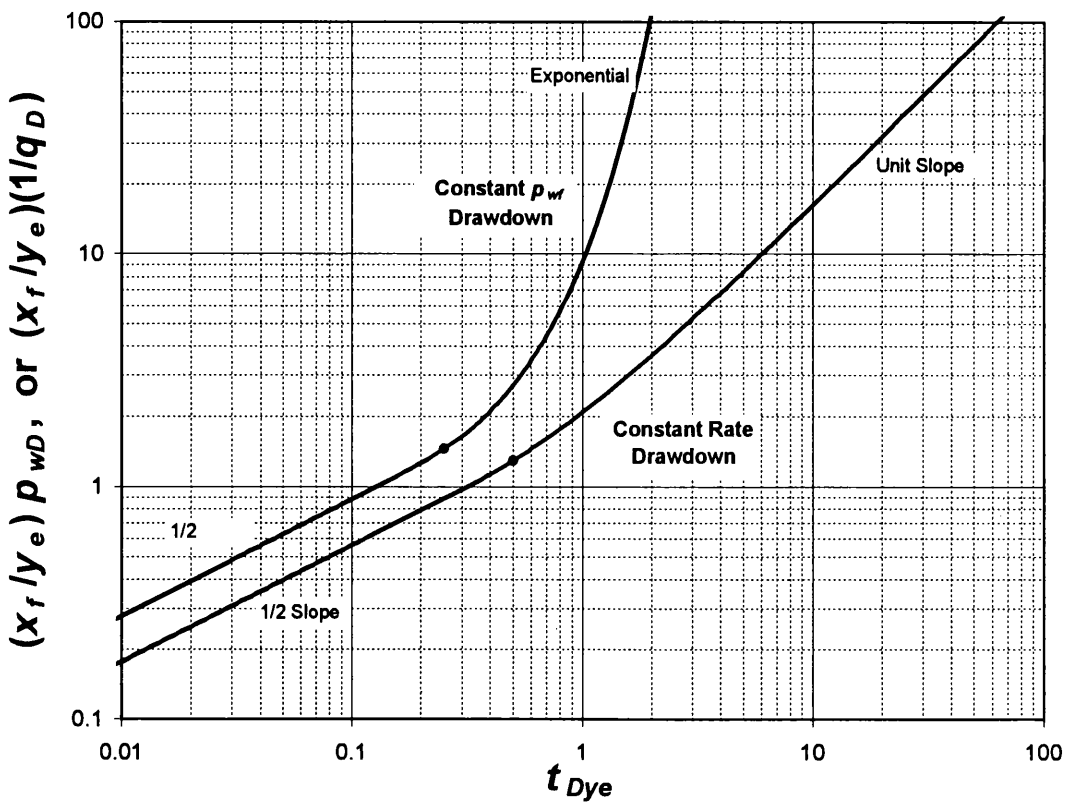

Fig. 2 - Constant rate and constant p_{wf} solutions for closed linear reservoirs.

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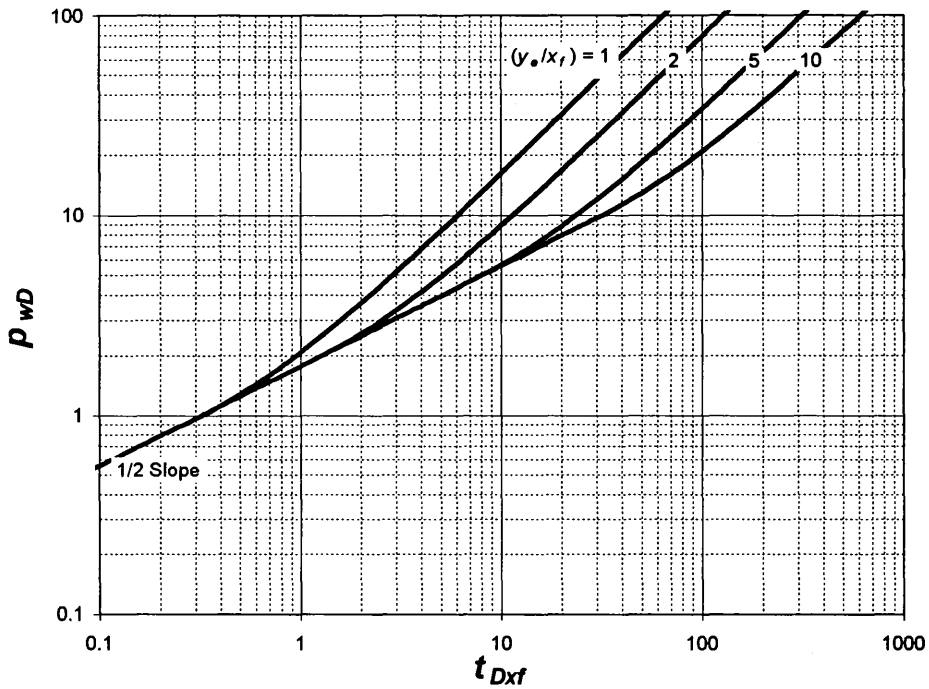

Fig. 3 - Constant production rate type curve for closed linear reservoirs.

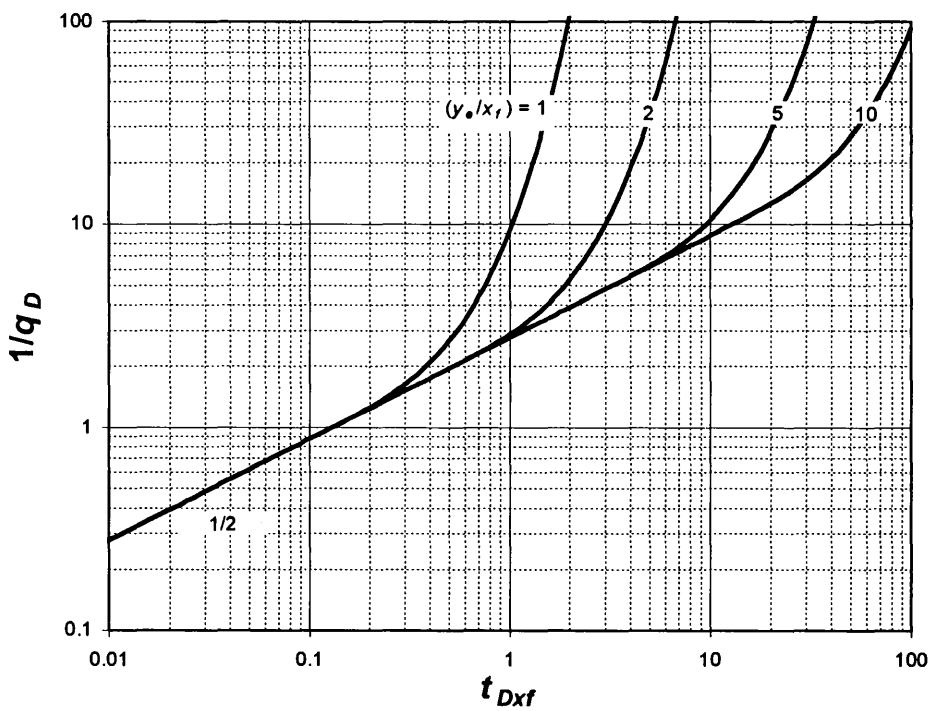

Fig. 4 - Constant p_{wf} type curve for closed linear reservoirs.

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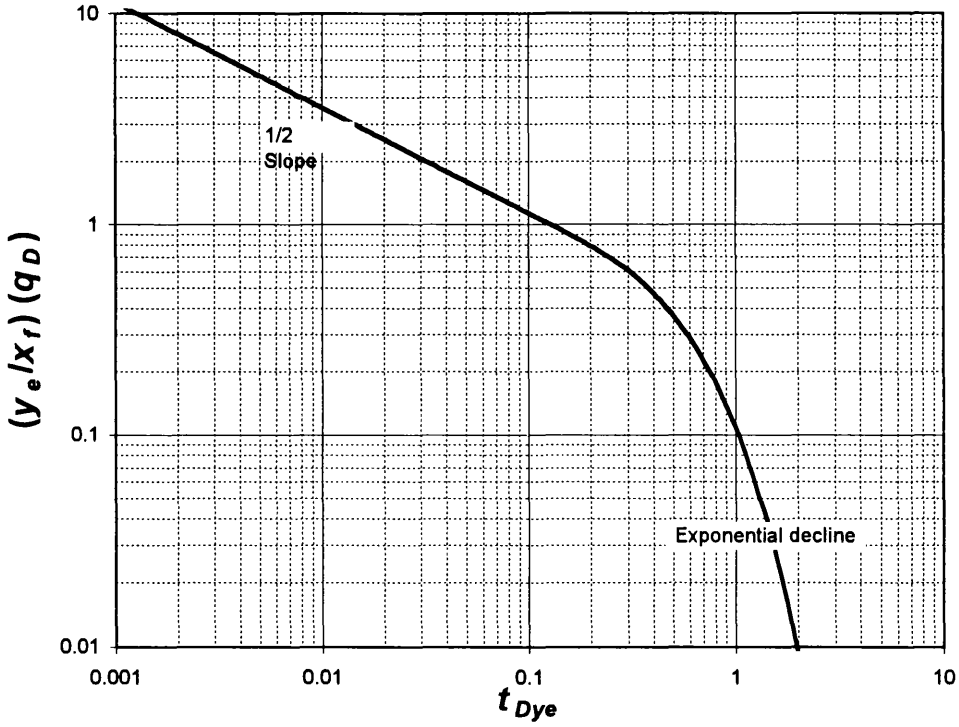

Fig. 5 - Decline curve for closed linear reservoirs.

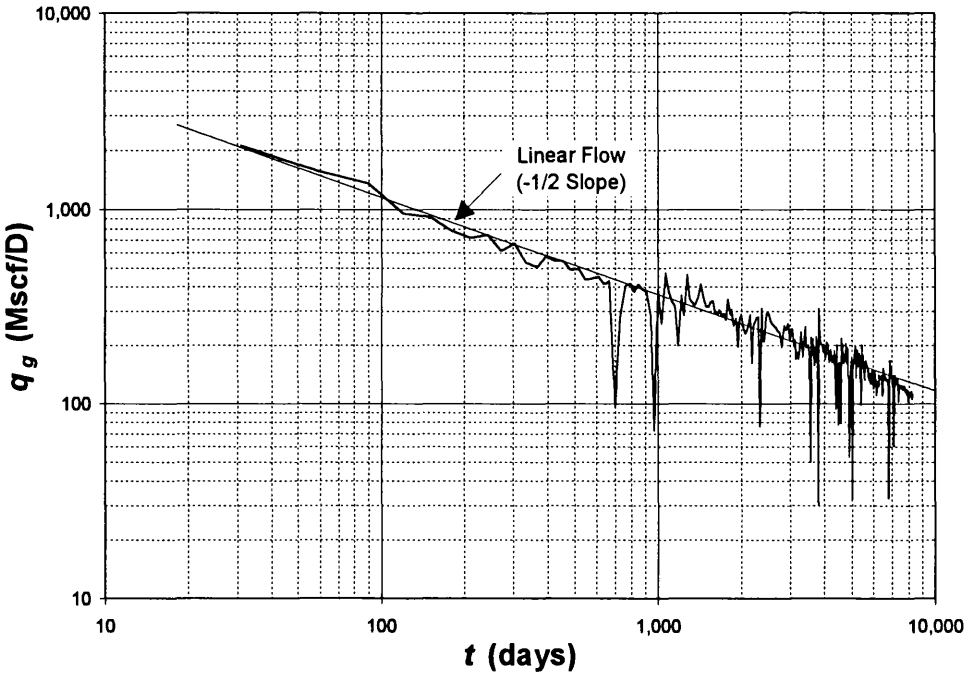

Fig. 6 - Log-log decline curve for example well.

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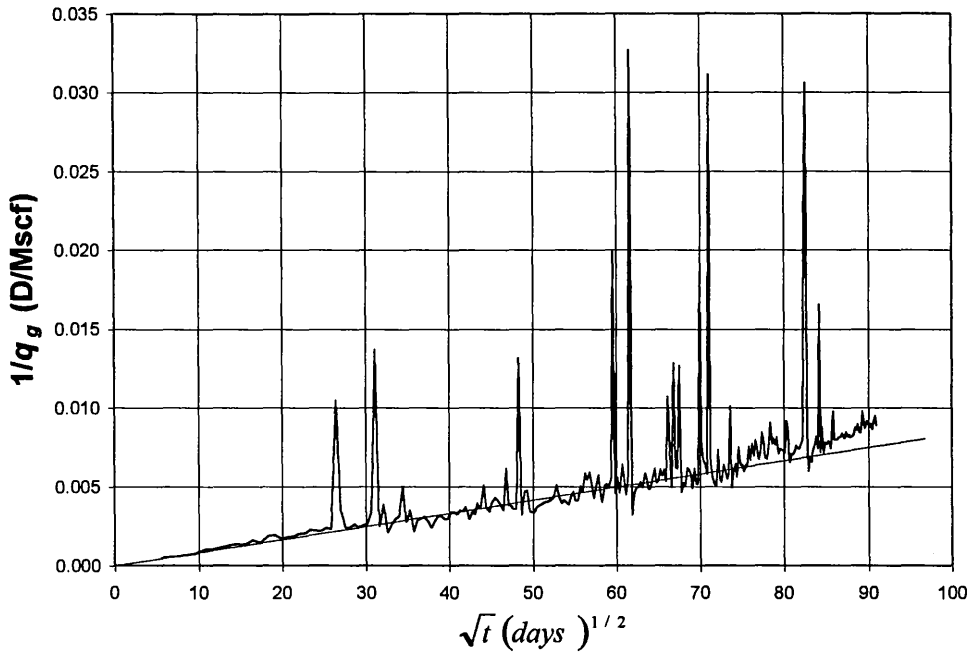

Fig. 7 - Square root of time plot for the reciprocal of rate of the example well.

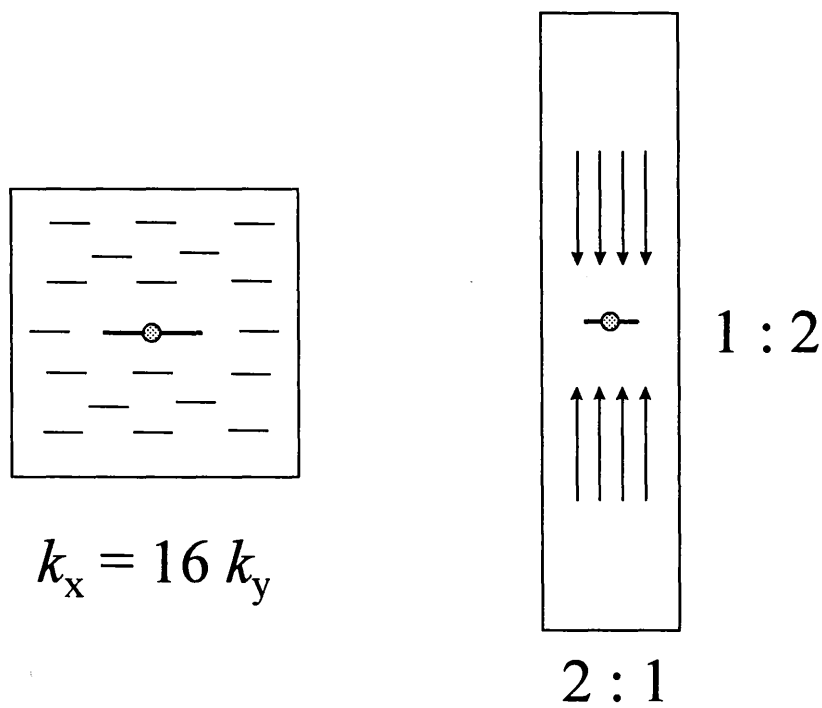

Fig. 8 - Effect of anisotropy from natural fractures, a square with k_x = 16 k_y transforms to a 4:1 rectangle.

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